# Full-Page Optical Music Recognition of Jazz Lead Sheets

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## Motivation & Contributions

OMR of **handwritten jazz lead sheets** requires transcribing both melody and chord symbols. Prior work relies on pre-segmented staff images, discarding page-level context.

**Research Questions:** *Can an end-to-end model directly transcribe a complete handwritten lead sheet and outperform a staff-level system—especially for chord-symbol transcription?*

To answer this, we contribute:

- the first **end-to-end full-page handwritten OMR** for jazz lead sheets, trained with a curriculum learning strategy.
- an **extended set of chord-oriented evaluation metrics** to enable a finer understanding of model capabilities and error types.

## Methodology

We adopt the state-of-the-art OMR architecture, **Sheet Music Transformer (SMT)**, and add a `<linebreak>` token to its vocabulary, then bridge staff-level pretraining to full-page recognition with 2 strategies of **9-stage curriculum training**:

- **Masking:** at stage $k$, vertically mask a real page in the training set so only the top $k$ staves remain visible, with the ground truth truncated.
- **Stacking:** at stage $k$, synthesize a multi-staff training set by stacking $k$ individual staff crops, with their per-staff annotations joined by `<linebreak>`.

**Experience replay** at each stage retains a fraction $r$ of earlier-stage samples to prevent forgetting.

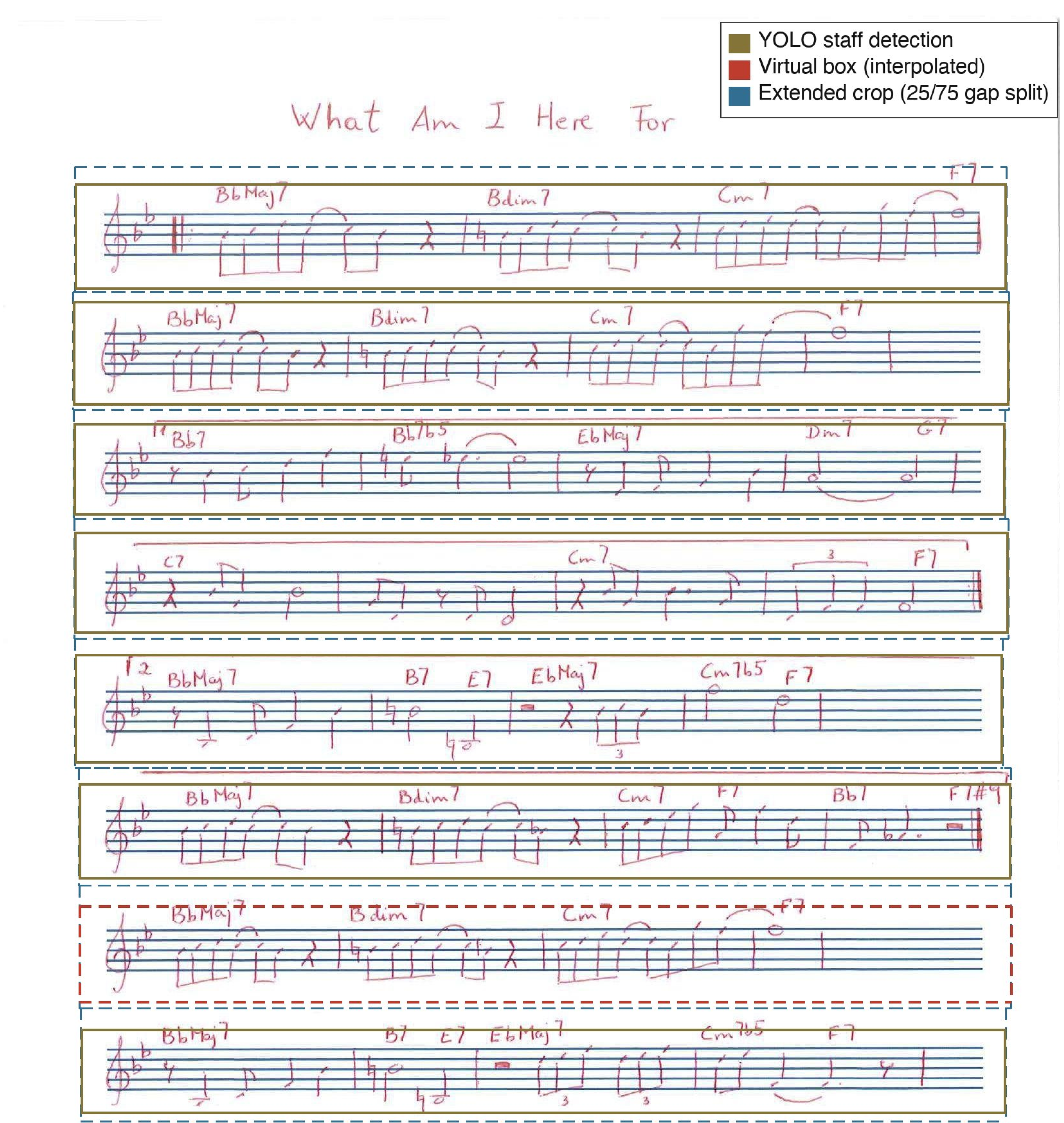
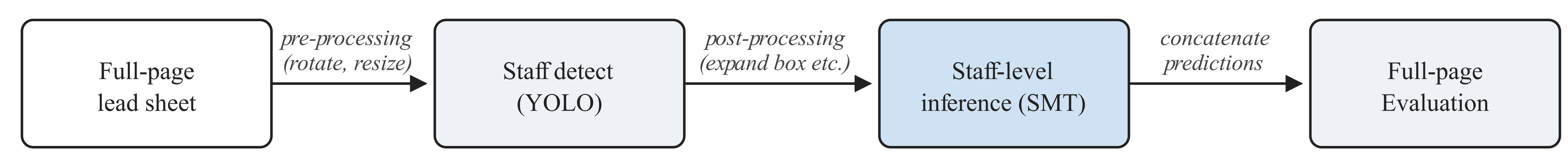



Figure 1. Sample of baseline staff detection and processing

## Baseline

As the first full-page handwritten jazz lead sheet OMR system, our model has no direct prior comparison. To enable a fair page-level evaluation, we construct a **detect-and-concatenate** baseline that lifts the existing staff-level recognizer to full pages:




## Evaluation Metrics

Standard edit-distance metrics (CER, SER, LER) treat melodic and harmonic errors equally. We isolate melodic accuracy by computing the standard metrics on the `**kern` spine only, and introduce two chord-oriented metrics on the `**mxhm` spine to evaluate harmonic transcription at different levels of strictness.

- $\mathrm{CER}_{\mathrm{Melo}}$, $\mathrm{SER}_{\mathrm{Melo}}$, $\mathrm{LER}_{\mathrm{Melo}}$: character/symbol/line edit distance computed on the `**kern` (melodic) spine only.
- $\mathrm{SER}_{\mathrm{Root}}$: edit distance over chord-root tokens only.
- $\mathrm{SER}_{\mathrm{Chord}}$: edit distance over the full chord symbol (root, quality, extensions, inversion).

## Results

Masking with full replay ($r$=1) outperforms the baseline on every overall metric; stacking underperforms.

| Method | CER | SER | LER |
|---|---|---|---|
| Baseline | 13.55 | 12.78 | 32.35 |
| Masking, $r = 0$ | 17.61 | 16.80 | 31.41 |
| Masking, $r = 0.25$ | 17.48 | 16.73 | 29.13 |
| Masking, $r = 0.50$ | 14.11 | 13.59 | 26.25 |
| Masking, $r = 1.00$ | **13.32** | **12.50** | **23.15**$^*$ |
| Stacking, $r = 0.25$ | 24.30 | 22.90 | 44.55 |

Table 1. Overall full-page recognition results. $^*p < 0.01$ *vs. baseline.*

| Metric | Base | Mask $r$=1 | $\Delta$ |
|---|---|---|---|
| $\mathrm{CER}_{\mathrm{Melo}}$ | 11.96 | **11.63** | $-0.33$ |
| $\mathrm{SER}_{\mathrm{Melo}}$ | 14.40 | **13.02** | $-1.38$ |
| $\mathrm{LER}_{\mathrm{Melo}}$ | 26.66 | **18.67**$^*$ | $-7.99$ |
| $\mathrm{SER}_{\mathrm{Root}}$ | 29.17 | **21.25**$^*$ | $-7.92$ |
| $\mathrm{SER}_{\mathrm{Chord}}$ | 47.82 | **30.67**$^{**}$ | $-17.15$ |

Table 2. Spine-specific metrics across masking replay ratios. $^*p < 0.01$, $^{**}p < 0.001$ *vs. baseline.*

## Discussion

**Key finding:** Full-page modeling improves chord recognition and line error rate.

- **Specific Improvements:** Common handwritten confusions (e.g., C–G, B–D, Bb–Eb) are substantially reduced.
- **Global Improvements:** Improvements are consistent across positions, not only near `<linebreak>` regions.
- **More context helps:** Gains are largest on dense pages ($N \geq 7$ staves).

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